

Anaya Koirala

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Education

Gettysburg College — B.S Computer Science and B.A Philosophy

GPA: 3.870

May 2028

Professional Experience

Software Tutor - Mathematics Department, Gettysburg College

Jan 2026–Present

- Tutor students in tools and softwares for quantitative courses across campus.
- Aid with debugging and problem solving in code based coursework.
- Specialized in Python, LaTeX, Java, WolframLang and statistical softwares.

Teaching Assistant for CS I and II — Computer Science Department, Gettysburg College

Jan–Dec 2025; Aug 2026–Present

- Supported approx. 30 students on core programming concepts, OOP, GUI, Data Structures
- Conducted weekly office hours for debugging, coursework, and technical issues

Software Developer — Kha Esports, Kathmandu, Nepal

Jun 2025–Aug 2025

- Developed website for hosting, managing, and participating in e-sports tournaments
- Worked on a multiplayer ludo game with for 64 simultaneous players
- Tools Used: FastAPI, MongoDB, React.js, Express.js, nginx

Relevant Coursework

- **Mathematics of Computation:** Combinatorics, Proofs, Graph Theory, Logic.
- **Calculus III:** Multivariable and Vector Calculus with Applications.
- **Linear Algebra:** Applications in compression, data analytics, Markov chains, IFS.
- **Artificial Intelligence:** Machine Learning, Reinforcement Learning, Game-Tree Search, Bayesian Reasoning, Markov Chains.

Technologies

Languages: Python, Java, Rust, C#, OCaml, F#, Shell, SQL, Haskell

Tools: Linux, BSDs, Windows, LaTeX, JUnit, NumPy, Sage, JavaFX

Projects

SubHunt — Audio-based PvE video game for blind players built in Godot

[GitHub](#)

- Worked in game design and development for the [Games for Blind Gamers 4](#) game jam.

MelodyTron — CS 371: Intro to AI Final Project

- Generate folk songs from an initial seed melody using latest machine learning tools.

[Website](#)

Musical-Style-Replicating-LSTM — IBDP Extended Essay project

[GitHub](#)

- Self-directed highschool research project in AI generated music emulating a composer's musical style.
- Achieved Grade 'A' using Python, LSTM-RNNs, TensorFlow, and NumPy

ireu - A classifier and database of REU programs that accept international students

[Website](#)

- Built in F# and Python using Web Scraping and LLM-based tools.

codeDP — Desktop code editor for IB Computer Science pseudocode

[GitHub](#)

- Built using Python and QT6

Academic Work

- "Is There a Collective in *Zhuangzi's* Freedom?" — MCCT Pre-Conference Student Panel 2026, Gettysburg College

Apr 2026

Awards

- [Andrew Marsh Memorial Award](#) — Gettysburg College

Oct 2025

Extra and Co-Curricular Activities

- **ACM Club** — Webmaster Team and Outreach Chair

Oct 2025 – Present

- **The Gettysburg Cinematics** — Senate Representative

Sept 2025 – Present

- **Quiz Bowl Club** — Senate Representative

Apr 2024 – Present

- **Amnesty International Club** — Treasurer

Sept 2025 – Present

- [The Gettysburgian](#) — Writer

Jan 2025– Present